# Volume of Cones, Pyramids and Spheres

Name _____ Date _____ Calculator allowed Version 1

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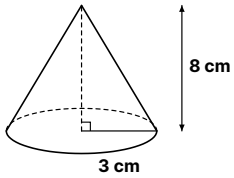
**KEY WORDS** cone *(koeko)* · pyramid *(koeko tapawhā)* · sphere *(pōro)* · hemisphere *(haurua pōro)* · apex

## A Cones ●○○

**Remember:** A cone is one third of the cylinder it fits inside: Volume $= \frac{1}{3}\pi r^2 h$. Use the perpendicular height, not the slant. Give answers to 1 decimal place.

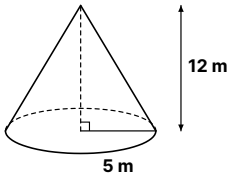
**1.** Find the volume of each cone.

**a)**



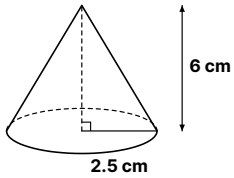

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**b)**



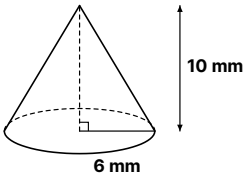

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**c)**



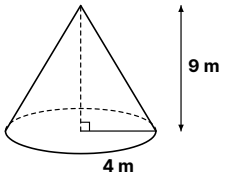

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**d)**



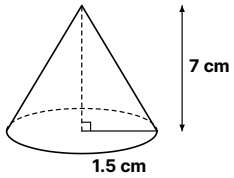

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**e)**




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**f)**



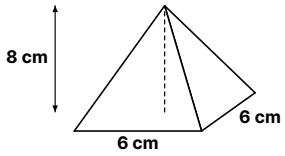

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## B Pyramids ●●○

**Remember:** A pyramid is one third of the prism with the same base and height: Volume $= \frac{1}{3}$ × base area × height.

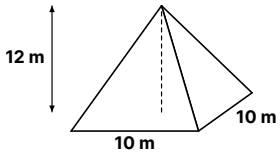
**2.** Find the volume of each pyramid.

**a)**



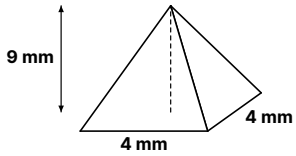

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**b)**



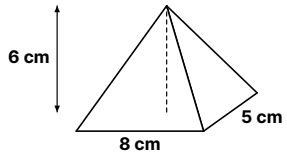

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**c)**



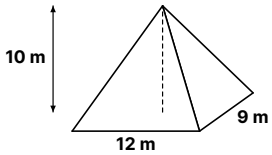

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**d)**



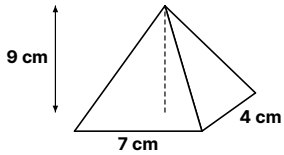

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**e)**




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**f)**




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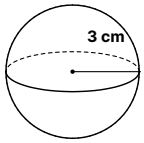
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## C Spheres and hemispheres ●●○

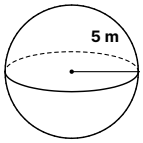
**Remember:** Volume of a sphere $= \frac{4}{3}\pi r^3$. Cube the radius first. A hemisphere is half of that.

**3.** Find the volume of each solid, to 1 decimal place.

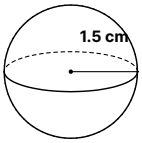
**a)**




**b)**




**c)**




**d)** A hemisphere (half a sphere) with radius 10 mm

**e)** A hemisphere (half a sphere) with radius 4 cm

**f)** A hemisphere (half a sphere) with radius 2.5 m

## D Working backwards ●●●

**Remember:** Put the volume into the formula and solve for the missing length. For a sphere you end with $r^3$: take the cube root.

**4.** Find the missing length.

**a)** A cone has a volume of 150 cm³ and a radius of 4 cm. Find its height, to 1 decimal place.

**b)** A square-based pyramid has a volume of 400 m³ and base edges of 10 m. Find its height.

**c)** A sphere has a volume of 500 cm³. Find its radius, to 1 decimal place.

**d)** A cone has a volume of 1,200 mm³ and a radius of 9 mm. Find its height, to 1 decimal place.

**e)** A square-based pyramid has a volume of 96 cm³ and base edges of 6 cm. Find its height.

**f)** A sphere has a volume of 3,000 mm³. Find its radius, to 1 decimal place.

## E Problems ●●●

**5.** An ice cream is a cone of radius 3 cm and height 10 cm, with a hemisphere of ice cream of the same radius on top. Find the total volume, to 1 decimal place.

**6.** Sand is tipped into a conical pile with a radius of 2.5 m and a height of 1.8 m. Find the volume of sand, to 2 decimal places.

**7.** A ball of diameter 20 cm is packed in a cube-shaped box with edges of 20 cm. Find the volume of empty space in the box, to the nearest cm³.

**8.** A roof is a square-based pyramid with base edges of 8 m and a height of 3 m. Find the volume of the roof space.

## ★ Extension

**9.** A cone, a hemisphere and a cylinder all have the same radius r, and the cone and cylinder have height r. Show that their volumes are in the ratio 1 : 2 : 3.

**10.** A sphere of radius 6 cm is melted and recast as a cone with the same radius. Find the height of the cone.



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# Volume of Cones, Pyramids and Spheres

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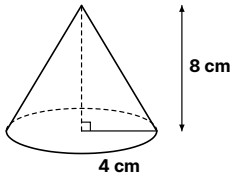
**KEY WORDS** cone *(koeko)* · pyramid *(koeko tapawhā)* · sphere *(pōro)* · hemisphere *(haurua pōro)* · apex

## A Cones ●○○

**Remember:** A cone is one third of the cylinder it fits inside: Volume $= \frac{1}{3}\pi r^2 h$. Use the perpendicular height, not the slant. Give answers to 1 decimal place.

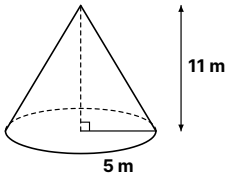
**1.** Find the volume of each cone.

**a)**



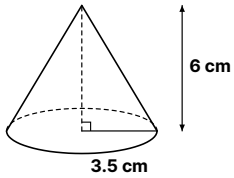

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**b)**



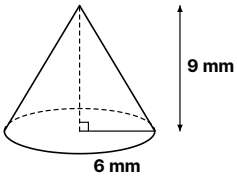

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**c)**



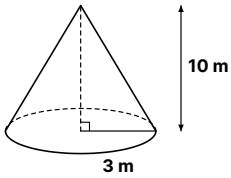

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**d)**



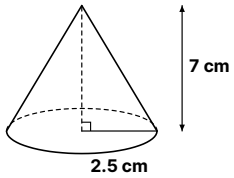

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**e)**




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**f)**



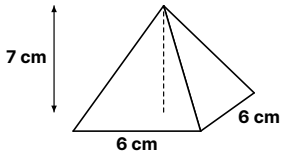

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## B Pyramids ●●○

**Remember:** A pyramid is one third of the prism with the same base and height: Volume $= \frac{1}{3}$ × base area × height.

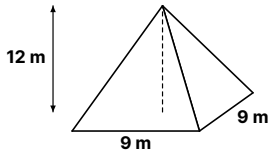
**2.** Find the volume of each pyramid.

**a)**



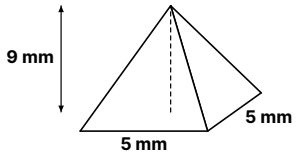

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**b)**



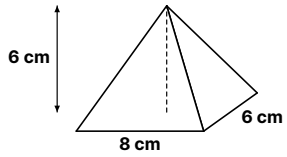

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**c)**



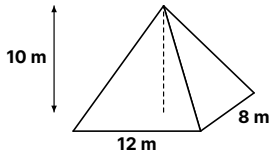

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**d)**



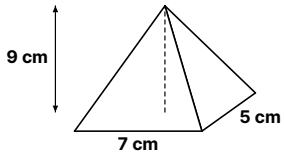

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**e)**




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**f)**




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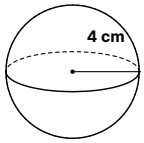
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## C Spheres and hemispheres ●●○

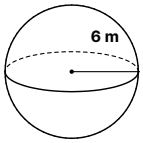
> **Remember:** Volume of a sphere $= \frac{4}{3}\pi r^3$. Cube the radius first. A hemisphere is half of that.

**3.** Find the volume of each solid, to 1 decimal place.

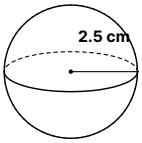
**a)**




**b)**




**c)**




**d)** A hemisphere (half a sphere) with radius 9 mm

**e)** A hemisphere (half a sphere) with radius 5 cm

**f)** A hemisphere (half a sphere) with radius 3.5 m

## D Working backwards ●●●

> **Remember:** Put the volume into the formula and solve for the missing length. For a sphere you end with $r^3$: take the cube root.

**4.** Find the missing length.

**a)** A cone has a volume of 200 cm³ and a radius of 5 cm. Find its height, to 1 decimal place.

**b)** A square-based pyramid has a volume of 324 m³ and base edges of 9 m. Find its height.

**c)** A sphere has a volume of 800 cm³. Find its radius, to 1 decimal place.

**d)** A cone has a volume of 1,500 mm³ and a radius of 10 mm. Find its height, to 1 decimal place.

**e)** A square-based pyramid has a volume of 128 cm³ and base edges of 8 cm. Find its height.

**f)** A sphere has a volume of 4,000 mm³. Find its radius, to 1 decimal place.

## E Problems ●●●

**5.** An ice cream is a cone of radius 3.5 cm and height 11 cm, with a hemisphere of ice cream of the same radius on top. Find the total volume, to 1 decimal place.

**6.** Sand is tipped into a conical pile with a radius of 3 m and a height of 2.1 m. Find the volume of sand, to 2 decimal places.

**7.** A ball of diameter 24 cm is packed in a cube-shaped box with edges of 24 cm. Find the volume of empty space in the box, to the nearest cm³.

**8.** A roof is a square-based pyramid with base edges of 9 m and a height of 4 m. Find the volume of the roof space.

## ★ Extension

**9.** A cone, a hemisphere and a cylinder all have the same radius r, and the cone and cylinder have height r. Show that their volumes are in the ratio 1 : 2 : 3.

**10.** A sphere of radius 9 cm is melted and recast as a cone with the same radius. Find the height of the cone.



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